

Toward a Better Understanding of Primary Care Physician Career Typologies

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Background: The nation faces a persistent shortage and maldistribution of primary care physicians (PCPs). A better understanding of PCP career typologies could help policy makers target interventions toward certain subgroups, rather than using a “one-size fits all” approach to improving PCP supply, distribution and diversity across settings and in rural areas.

Methods: This study used cross-sectional data from 2009 and 2019, derived from the North Carolina (NC) Medical Board, on PCPs in active practice in family medicine, general internal medicine, general pediatrics, geriatrics, and obstetrics and gynecology in NC. We used latent class analysis (LCA) to investigate: (1) whether different career typologies exist in the primary care physician workforce; (2) if so, whether career typologies changed in the 10-year period before the COVID-19 pandemic (2009 and 2019); and (3) whether a physician’s generational cohort, age, sex, race/ethnicity, career stage, and medical school location were associated with different career typologies.

Results: The LCA yielded 4 distinct career typologies in both 2009 and 2019 with high levels of class separation. The 4 typologies were relatively stable over the decade. Distinguishing factors between typologies included practice in a rural area, hospital, and ambulatory care employment, and provision of obstetric and prenatal care.

Conclusions: Understanding PCP career typologies could be used to tailor interventions to improve the supply and distribution of PCP workforce. Medical schools could use this information to support trainees’ decision-making about future careers and

policy makers to target funding for training to support careers in rural and ambulatory settings.

Key Words: careers, health workforce, latent class analysis, primary care, primary care physician

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Despite compelling evidence that access to high-quality primary care improves the health of individual patients and populations,^{1,2} ~101 million Americans currently live in primary care health professional shortage areas in the United States (US). Eliminating these shortages would require 8500 additional primary care clinicians.³ Fewer graduates from US medical schools are choosing primary care careers.^{4,5} Rural areas face significant primary care physician (PCP) shortages and an impending wave of physician retirements.⁶ Attracting new physicians to practice primary care in rural areas to replace those leaving the workforce remains challenging.^{7,8} In North Carolina (NC), for example, only 14% of medical students trained in-state are in a primary care specialty 5 years after graduation and only 1% are in rural primary care.⁹

Physicians from racial and ethnic groups that are underrepresented in medicine (URiM) disproportionately practice in underserved communities and in primary care.^{10–12} However, the growth of URiM PCPs has been minimal over time.¹³ The percentage of Black male medical students has declined over the past 3 decades^{14,15} and there has been little to no increase in Black or Hispanic men among physicians from 2000 to 2019.¹⁶

In contrast, the sex composition of the physician workforce has changed significantly. The number of female physicians has increased significantly in the US, growing from 28.3% of the workforce in 2007 to 36.3% in 2019¹⁷ and the proportion of PCPs who are women has nearly doubled in recent decades.¹⁸ Female physicians are more likely to pursue specialties such as general internal medicine, general pediatrics, family medicine, or obstetrics-gynecology.⁵ In 2021, women accounted for 65.0% of active physicians in pediatrics and 60.5% in obstetrics and gynecology.¹⁹

At the same time, the broader context of primary care practice has also undergone significant change in recent years. From 2016 to 2018, the percentage of PCPs

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affiliated with vertically integrated health systems rose from 38% to 49%, and there has been a corresponding decline in the proportion of physicians who are full or part owners of their practices.^{20,21}

Viewed holistically, these trends suggest that the forces affecting PCPs' careers are likely a complex interaction of individual, contextual, and time-dependent factors. Developing a better understanding of whether certain career typologies exist within the primary care workforce could help policymakers target interventions toward certain subgroups that may exist in the workforce. Drawing on a unique, longitudinal dataset of PCPs licensed to practice in North Carolina, this exploratory study used latent-class analysis to investigate: (1) whether different career typologies exist in the PCP workforce; (2) if so, whether career typologies changed in the 10-year period before the COVID-19 pandemic (2009 and 2019); and (3) whether a physician's generational cohort, age, sex, race/ethnicity, career stage, and medical school location were associated with different typologies.

CONCEPTUAL FRAMEWORK

This study draws on life course theory as a theoretical and methodological framework to understand how changing physician demographics intersect with the changing medical context to influence physician career typologies. Seminal work by Elder (1975; 2003) demonstrated how major social and historical events affect how people develop, behave, and age through their lives.^{22,23} Life course theory suggests that individual PCP's career behaviors—and aggregate trends in the PCP workforce—are shaped not only by physicians' individual decisions but also by the broader forces shaping their lives and the health care system during the period in which they are living.^{22,24} Prior life course studies have made significant contributions to our understanding of how individuals' careers unfold and the influence of age^{25,26} and sex²⁷ on career pathways, as well as the intersectionality of sex, and race on professions.²⁸ There is evidence that female physician's experiences in the medical workplace differ from their male counterparts²⁹ and that female physicians' attempts to combine family responsibilities with careers impacts their behaviors and choices.^{30,31} This study's hypotheses and methods draw on both life course and intersectionality research to investigate how a physician's age, sex, and race/ethnicity combine with historical forces to produce different career typologies.

METHODS

Data Source and Sample

This study used cross-sectional data from 2009 and 2019 derived from the NC Medical Board. The Medical Board collects self-reported information about physicians' demographic, practice, training, and geographic characteristics during initial licensure and annual licensure renewals. Because licensure is required to practice, the data represent a complete census of physicians in active clinical practice in NC in each year. Federally employed physicians,

physicians licensed in NC but working in another state, and residents in training were excluded from the sample.

Study Variables

To investigate the career typologies of PCPs, we chose indicator variables that captured key dimensions of medical practice.

Primary Care Specialty

The sample includes PCPs whose self-reported practice area was in family medicine, general internal medicine, general pediatrics, geriatrics, and obstetrics and gynecology (ob-gyn). Ob-gyns are included because they are considered PCPs by the NC General Assembly³² and because they provide prenatal, obstetric, and primary care services. In 2010, the NC Medical Board redesigned their annual licensure forms to move from asking physicians about their specialty to asking about their primary "area of practice" to capture more precise information about what they "normally do" in practice. Before this change, physicians who completed residency training in internal medicine and pediatrics but went on to subspecialize may have been categorized in general internal medicine or general pediatrics, instead of their subspecialty practice area. In addition, in 2011, physicians began to have the option of choosing "hospitalist" as a primary area of practice on the licensure form. Previously, family medicine physicians, internists, general pediatricians, geriatricians, and ob-gyns practicing as hospitalists would have been counted as PCPs. Therefore, although we excluded the 1425 physicians who reported they were hospitalists in 2019 from our sample, we could not identify hospitalists in 2009, and they are included in the PCP counts in that year. The exclusion of hospitalists from primary care in 2019 likely contributed to the decline in PCPs in NC from 8702 in 2009 to 7929 in 2019 (Table 1).

Employment Setting

Physicians' primary employment settings were coded in 6 categories: solo practitioner's office, ambulatory care (free-standing clinic, group office, and staff or group model HMO), hospital (outpatient, inpatient, emergency room, or other hospital setting), medical school or parent university, nursing home/extended care facility, other (locum tenens, home health, telemedicine, volunteer, hospice care, and other).

Obstetrics and Prenatal Care

Physicians were asked "are you providing obstetric deliveries as a routine part of your practice and are you providing prenatal care as a routine part of your practice?"

Hours Worked Per Week

Physicians report, for all practice locations held in NC, the average number of hours per week spent in patient care, including charting and image reading.

Rural/Urban Practice

We geocoded physician practice addresses and assigned them to Rural-Urban Commuting Area (RUCA)

TABLE 1. Primary Care Physicians in North Carolina, 2009 and 2019

Variables	2009 (n = 8702)	2019 (n = 7929)	Sign.
Specialty (%)			
Family medicine	33	36	***
General internal medicine	35	26	***
General obstetrics and gynecology	13	15	*
General pediatrics	19	21	*
Geriatrics	0	3	***
Employment setting (%)†			
Ambulatory care	53	54	
Hospital	20	18	*
Medical school/parent university	7	3	***
Nursing home/extended care	1	2	***
Other	1	9	***
Solo practitioner	11	7	***
Practice characteristics (%)			
Providing routine obstetric deliveries	11	12	***
Providing routine prenatal care	14	15	*
Avg patient-care hours per week	40.8	38.4	***
Urban/rural status (%)			
Urban	80	83	***
Large rural	14	12	*
Small rural	6	5	***
Female (%)	39	50	***
Racial/ethnic identity (%)‡			
Non-Hispanic White	72	67	***
Non-Hispanic Black	11	11	
American Indian/Alaskan Native	1	1	
Asian/Pacific Islander	10	11	
Hispanic	3	4	
Other	3	3	
Generational cohort (%)			
Gen Y (1981–2000)	0	21	***
Gen X (1965–1980)	44	42	*
Baby boomer, trailing edge (1955–1964)	28	23	***
Baby boomer, leading edge (1946–1954)	20	11	***
WWII (1928–1945)	8	3	***
Average years of practice (STD)	19.97 (11.0)	22.5 (12.4)	***
Average age (STD)	46.5 (10.8)	49.2 (12.2)	***
Medical school (%)			
IMG	17	15	*

†In 2009, 6% of physicians were missing setting and in 2019, 7% of PCPs were missing setting.

‡In 2019, 3% were missing race/ethnicity.

* $P < 0.05$.

*** $P < 0.001$.

codes³³ that identify practice addresses as urban, large rural, and small rural categories.

Demographic and Educational Characteristics

Average age was calculated, and race/ethnicity categories included non-Hispanic White, non-Hispanic Black, American Indian/Alaskan Native, Asian/Pacific Islander, Hispanic, and other race/ethnicity. A physician's generational cohort was derived based on their birth year and included physicians in generation Y (birth years 1981–2000), generation X born (1965–1980), trailing edge baby boomer (1955–1964), leading edge baby boomer

(1946–1954), and WWII (1928–1945). Medical school cohorts were constructed based on years since medical school graduation (0–5 y, 6–10, 11–20, 21–30, and 31 or more). The analysis coded physicians as US-educated versus international medical graduates (IMGs) with the latter category including both US- and foreign-born physicians.

Statistical Analysis

Descriptive statistics were used to report PCP specialty, employment setting, provision of obstetric and prenatal care, average patient-care hours per week, and rural practice location. We then used latent-class analysis (LCA) to assess the presence of unobserved subgroups in the sample that possessed similar response patterns with respect to 4 domains, representing key dimensions of a physician's practice—practice in a rural versus urban location, specialty, employment setting, and provision of obstetric and prenatal care services.³⁴ To identify an optimal solution, we evaluated relative fit using pre-determined criteria: Akaike information criterion (AIC), Bayesian information criterion (BIC), adjusted BIC (aBIC), likelihood ratio tests (including the Lo-Mendell-Rubin [LMR] likelihood ratio test, the adjusted LMR test, and the bootstrap likelihood ratio test), mean posterior probabilities, entropy, and substantive interpretability.^{35,36} Together, these criteria signal which latent-class solution best fits the data and creates the highest levels of classification certainty. We conducted 2 separate LCAs for 2009 and 2010 using the same indicators.

After selecting the optimal solution, we used the 3-step procedure³⁷ to estimate sociodemographic covariate differences across latent classes. The 3-step procedure aids in accounting for classification uncertainty by simultaneously extracting latent classes, assigning respondents to their most likely class, and assessing covariate differences across classes. LCA was conducted in Mplus 8.6 and preliminary data management was conducted in Stata 17.0.

RESULTS

Table 1 describes the characteristics of PCPs practicing in NC in 2009 and 2019. The overall number of PCPs declined and there were fewer internal medicine physicians in 2019, likely due to the change in how specialty was captured on the licensure form between 2009 and 2019. There was an increase in geriatricians between 2009 and 2019. The percentage of PCPs practicing in medical schools declined from 7% in 2009 to 3% in 2019 and solo practice PCPs declined from 11% to 7% of PCPs in the same time period. The “other setting” setting category increased from 1% to 9%. There were significant differences in the sociodemographic characteristics of PCPs in NC between 2009 and 2019, which suggests the importance of using life course theory as lens for understanding how a physician's age, sex, and generational cohort may influence practice behaviors. The average age of the workforce increased from 46.5 to 49.2, and female PCPs increased from 39% of the workforce in 2009 to 50%

in 2019, gen Y was not yet present in the workforce in 2009 but made up 21% of PCPs in NC in 2019. There were more non-Hispanic White physicians in 2009 than 2019 yet no other racial/ethnic differences between 2009 and 2019.

Latent-Class Enumeration

Table 2 shows the LCA yielded 4 distinct career typologies or “classes” (we use these terms interchangeably in this manuscript), in both 2009 and 2019 with high levels of class separation and classification certainty (entropy = 0.98, mean posterior probabilities > 0.90). The key variables that distinguished between the classes were whether the PCP practiced in a rural or urban area, was used in a hospital or ambulatory setting, and whether they provided obstetric and prenatal care. We named the classes based on these distinguishing factors.

Latent-Class Description

Table 3 shows the 4-class solution in 2009. Class 1 (“Urban Internists”) represented 28% (n = 2413) of NC’s PCPs and was comprised completely of general internists working in urban areas across multiple employment settings. Class 2 (“Urban PCPs”) was the largest class, making up 41% (n = 3573) of the 2009 PCP workforce. No internists were in class 2; instead, the group was comprised mostly of family medicine physicians and pediatricians working across multiple settings in urban areas. Class 3 (“Rural PCPs”) comprised 20% (n = 1772) of NC’s primary care workforce and included a mix of family medicine physicians, general internists, pediatricians, and a small proportion of ob-gyns practicing in small and large rural areas. Nearly one-fifth (18%) of rural PCPs were in solo practice. Class 4 [“Ob-gyns and Family Medicine (FM) Providing Obstetric and Prenatal Care”] was the smallest class, representing 11% of the sample (n = 944). The majority (89%) of Class 4 reported they were ob-gyns, yet 12% were family medicine physicians. More than two-thirds (70%) practiced in ambulatory settings and 80% of physicians in class 4 were in urban areas.

Table 4 shows the 4-class solution in 2019. Class 1 (“Urban Non-Ambulatory Care PCPs”) comprised 32%

(n = 2569) of the sample and had more general internists practicing in hospitals than other classes, and all were practicing in urban areas. None of the physicians in class 1 practiced in ambulatory care (defined as free-standing clinics, group offices, and staff or group model HMO practices) and they were more likely than other classes to practice in hospitals (40% of class 1), medical schools (8%), and solo practitioner offices (15%). Class 2 (“Urban ambulatory care PCPs”) was 37% (n = 2940) of the sample with a mix of primary care physician specialties, all practicing in ambulatory care in urban areas. None of the physicians in class 1 or class 2 provided obstetric deliveries. Class 3 (“Rural PCPs”) in 2019 represented 18% (n = 1425) of the sample with nearly one-half (48%) of the class made up of family medicine and the rest were a mix of other primary care specialties. Class 3 practiced in a variety of settings in small and large rural places. Class 4 (“Ob-gyns and FM Providing Obstetric and Prenatal Care”) made up 13% (n = 955) of the PCP workforce in 2019. Ob-gyns comprised the majority (88%) of class 4 (family medicine physicians comprised the remainder), providing obstetric and prenatal care services across settings including hospitals, ambulatory care, medical schools, and other settings. Given that North Carolina’s legislative definition includes ob-gyns as primary care clinicians, we conducted a sensitivity analysis to assess potential changes in our findings if we removed them from the LCA. We found that although Class 4 disappeared when ob-gyns were not included, the LCA yielded a 3-class solution in which classes 1, 2, and 3 looked very similar to when we included ob-gyns in the analysis.

Table 5 shows associations between latent-class membership and sex, race/ethnicity, generational cohort, years in practice, age, and IMG status. In both time periods, females comprised a smaller percentage of the rural PCP class and a larger proportion of PCPs providing obstetric and prenatal care (class 4). In both 2009 and 2019, the PCP workforce in class 4 was less diverse and PCPs in class 1 were the most racially and ethnically diverse. Consistent with the literature,^{10–12} PCPs in rural areas (class 3) in both years were older and had more years in practice, on average, than other classes. Between

TABLE 2. Latent Class Analysis

Classes	Log likelihood	Parameters	Solution replicated	AIC	BIC	aBIC	P			Entropy	Smallest class size
							LRT	aLRT	BS LRT		
1	−112,479	24	NA	225,007	225,176	225,100	NA	NA	NA	NA	NA
2	−107,579	47	Yes	215,251	215,584	215,435	0	0	0	0.96	1993
3	−103,525	70	Yes	207,189	207,684	207,462	0	0	0	0.97	946
4	−100,916	93	Yes	202,017	202,675	202,379	0	0	0	0.98	944
5			No								
6			No								
1	−102,998	24	NA	206,044	206,212	206,136	NA	NA	NA	NA	NA
2	−98,434	47	Yes	196,963	197,291	197,141	0	0	0	0.99	1025
3	−94,992	70	Yes	190,123	190,612	190,389	0	0	0	0.96	1004
4	−92,549	93	Yes	185,285	185,934	185,638	0	0	0	0.98	995
5			No								
6			No								

TABLE 3. Item Response Probabilities for Each Indicator in the 4-Class Model, 2009

Indicators	1—Urban internists (n = 2413, 28%)	2—Urban PCPs, no internists (n = 3573, 41%)	3—Rural PCPs (n = 1772, 20%)	4—Ob-gyns and FM providing obstetric and prenatal care (n = 944, 11%)
Family medicine (%)	0	53	45	12
General internal medicine (%)	100	0	37	0
General obstetrics and gynecology (%)	0	7	3	89
General pediatrics (%)	0	38	15	0
Geriatrics (%)	0	1	0	0
Ambulatory care (%)	39	58	53	70
Hospital (%)	31	15	21	9
Medical school/parent university (%)	12	8	0	7
Nursing home/extended care (%)	1	1	1	0
Solo practitioner (%)	9	10	18	8
Other setting (%)	2	1	2	1
Provides routine obstetric deliveries (%)	0	0	0	96
Provides routine prenatal care (%)	1	5	3	98
Urban (%)	100	100	16	80
Large rural (%)	0	0	59	15
Small rural (%)	0	0	25	6

2009 and 2019, the average age of the PCP workforce in class 3 increased from 48.2 to 52.8. Gen Y, who entered the workforce between 2009 and 2019, were more likely to practice in urban, nonambulatory care (class 1) and provide obstetric and prenatal care (class 4). Gen X physicians were evenly distributed across classes in both years but more likely to also be found in class 4 in 2009 before Gen Y entered the workforce. Leading edge boomer PCPs (born 1946–1954) made up a higher per-

cent of rural PCPs. IMGs made up the greatest proportion of class 1 (urban PCPs more likely to be employed in hospitals) and class 3 (rural PCPs) in both 2009 and 2019.

DISCUSSION

To our knowledge, this is the first latent-class analysis to identify career typologies within the physician

TABLE 4. Item Response Probabilities for Each Indicator in the 4-Class Model, 2019

Indicators	1—Urban, nonambulatory care PCPs (n = 2569, 32%)	2—Urban ambulatory care PCPs (n = 2940, 37%)	3—Rural PCPs (n = 1425, 18%)	4—Ob-gyns and FM providing prenatal and obstetric care (n = 995, 13%)
Family medicine (%)	33	40	48	12
General internal medicine (%)	37	25	26	0
General obstetrics and gynecology (%)	5	4	4	88
General pediatrics (%)	19	29	20	0
Geriatrics (%)	7	1	3	0
Ambulatory care (%)	0	100	52	63
Hospital (%)	40	0	16	20
Medical school/parent university (%)	8	0	1	4
Nursing home/extended care (%)	4	0	1	0
Solo practitioner (%)	15	0	12	2
Other setting (%)	17	0	13	5
Provides routine obstetric deliveries (%)	0	0	1	98
Provides routine prenatal care (%)	5	2	4	97
Urban (%)	100	100	21	84
Large rural (%)	0	0	56	13
Small rural (%)	0	0	23	3

TABLE 5. Characteristics of Primary Care Physician Typologies, 2009 and 2019

Variables	2009					2019 sample				
	Sample mean (N = 8702)	Class 1: Urban internists in multiple settings (n = 2413; 28%)	Class 2: Urban PCPs, no internists (n = 3573; 41%)	Class 3: Rural PCPs (n = 1772; 20%) (%)	Class 4: Ob/gyns and family medicine providing ob/prenatal care (n = 944; 11%)	Sample mean (n = 7929)	Class 1: Urban nonambulatory care PCPs (n = 2569; 32%)	Class 2: Urban ambulatory care PCPs (n = 2940; 37%)	Class 3: rural PCPs (n = 1425; 18%)	Class 4: Ob-gyns and family medicine providing ob/prenatal care (n = 995; 13%)
Female (%)	39	34	45	33	45	50	52	50	40	62
Racial/ethnic identity (%)										
Non-Hispanic White	72	64	78	66	80	67	60	73	66	73
Non-Hispanic Black	11	12	9	12	12	11	12	9	13	12
American Indian/Alaskan Native	1	0	0	2	1	1	0	0	2	1
Asian/Pacific Islander	10	15	8	12	4	11	15	10	9	5
Hispanic	3	3	3	5	3	4	5	3	4	3
Other	3	5	2	4	1	3	3	3	3	2
Missing Race	< 1	< 1	< 1	< 1	< 1	3	5	2	3	4
Generational cohort (%)										
Gen Y (1981–2000)	0	0	0	0	0	21	28	16	14	32
Gen X (1965–1980)	44	47	44	38	52	42	37	48	37	45
Baby Boomer, Trailing Edge (1955–1964)	28	29	28	28	28	23	20	24	27	17
Baby Boomer, Leading Edge (1946–1954)	20	16	20	24	16	11	12	10	18	5
WWII (1928–1945)	8	8	8	10	5	3	4	2	4	1
Years in practice	19.97 (11.0)	19.6 (10.9)	20.03 (11.0)	21.28 (11.3)	18.15 (9.8)	22.5 (12.4)	21.5 (13.6)	23.3 (11.2)	25.6 (12.6)	18.2 (11.0)
Age	46.5 (10.8)	45.7 (10.8)	46.6 (10.8)	48.2 (11.1)	44.9 (9.8)	49.2 (12.2)	48.2 (13.4)	49.8 (10.9)	52.8 (12.3)	45.0 (11.0)
Medical school (%)										
IMG	17	24	11	26	4	15	20	13	20	4

Pairwise class differences at $P \leq 0.05$ were observed for all variables within both 2009 and 2019, except for gen Y in 2009.

workforce. The LCA sorted PCPs into 4 career typologies in 2009 and 2019 with high levels of class separation and classification certainty. This suggests that the dimensions of practice included in the LCA—specialty, employment setting, practice in rural versus metropolitan counties, and provision of obstetric and prenatal care—were well-suited to differentiate between distinct career typologies of PCPs in North Carolina.

The stability of PCP career typologies between 2009 and 2019 was surprising. We hypothesized that changes in the demographic composition of the workforce, the rapid pace of health care system change, and differences in how physicians reported their specialty in 2009 versus 2019 would produce different career typologies in the 2 periods. However, career typologies were not only similar in both years, but the percentage of the workforce falling into each career type was also relatively stable. Classes 1 and 2 were less stable between the 2 periods, likely due to the change in how specialty information was collected and the fact that 1425 hospitalists were excluded from the 2019 data. In both 2009 and 2019, about 20% of PCPs were in the rural PCP class whose membership consisted of older, more racially and ethnically diverse family physicians and pediatricians practicing in ambulatory care settings in rural areas. A higher proportion of these physicians were IMGs who, through the J-1 visa waiver program, can bypass the requirement to return to their home country after residency by practicing in rural communities. PCPs in Class 4 were also consistent between the 2 periods, making up about 11%–13% of the PCP workforce and comprised of obstetricians/gynecologists and family medicine physicians providing obstetric and prenatal care and practicing predominantly in ambulatory settings in urban areas. This group was younger, more female, less racially/ethnically diverse, and less likely to be IMGs than other classes in both years.

If we assume that these 4 career typologies are relatively immutable and will remain the dominant subgroups within primary care, the question becomes how to use this knowledge to improve the supply, distribution, and diversity of the primary care workforce.

Policymakers could use the knowledge that PCP careers are relatively “fixed” over time to target investments to better align with supporting different career typologies. For example, fewer than 20% PCPs in both years practiced in hospital-based settings. Yet, most Medicare funding for graduate medical education (GME) flows to hospitals and only 2% of GME training occurs in rural areas.^{38,39} These data align with other work⁴⁰ that suggests that federal policymakers should support the PCP workforce and continue to invest in training models in community-based ambulatory practice in underserved and rural communities.

At the state level, Medicaid and state appropriations are an important and growing source of GME funds and state policymakers could use these data to target investments toward needed specialties, geographies, and settings.⁴¹ Recent work by Patterson et al⁴² underscores the importance of rural residency training as a strategy to in-

crease the number of physicians practicing in rural communities. That study supports a “grow your own” approach that recruits physicians from rural backgrounds into rural residencies and shows that the combination of rural background and rural residency training yields the highest probability of future practice in rural area. The current study found that the rural workforce was the oldest class in 2019 at an average age of 52.8, up from 48.2 in 2009. This finding raises the important question of whether new entrants will move to rural areas to replace the retiring workforce or if physicians in other classes will move to these areas later in their career.

Medical schools vary considerably in their contribution to the social mission of medical education, measured in part by the proportion of their graduates who go on to practice primary care and work in underserved areas.⁴³ Previous work has shown that NC’s medical schools have different “footprints” in terms of the degree to which they produce primary care and rural physicians.⁴³ This analysis did not investigate medical schools representation among the 4 different typologies but such an analysis would be a valuable extension of this study and could guide decisions to invest in medical schools that produce the primary care physician workforce needed to meet population health needs.

LIMITATIONS

The generalizability of findings from this study may be limited as NC PCPs’ demographic and practice characteristics and medical practice context differ nationally or from other individual states. State policies that may shape practice were not examined in this study. For example, we did not have access to data on which physicians in this study were recipients of state and federal loan repayment programs, which could have, for example, been more likely for class 3, the rural physician group.

States could replicate this analysis using licensure or other datasets that contain the similar demographic, practice, and geographic variables used in this study. This would be a valuable analysis to help state and federal policymakers understand if primary care career typologies differ between states, and if so, how state workforce policies might be specifically targeted to shape the size and composition of the workforce. If typologies are similar across states, this would suggest that federal policy efforts to recruit, retain, and sustain the PCP workforce may be helpful to augment states’ efforts to address primary care shortfalls.

An important limitation of this study is that the LCA was conducted using data on primary care physicians only. Advanced practice registered nurses (APRNs) and physician assistants (PAs) make up 39% of the primary care workforce in North Carolina between 2017 and 2019 and comprise a large proportion of the primary care workforce in the state’s most underserved counties.⁴⁴ National data reflect these trends; between 2013 and 2019, the proportion of Medicare visits delivered by either a Nurse Practitioner (NP) or PA increased from 14.0% to 25.6%.⁴⁵ Taken together, these trends suggest that NPs

and PAs will continue to play an important and growing role in primary care in the United States in the future. Future analyses could use LCA to explore whether there are unobserved patterns of clusters among PCPs, and primary care APRNs and PAs who practice together in different settings and geographies.

This study analyzed physician careers before the pandemic and there is reason to believe that COVID-19 disrupted careers in ways not seen in the past.^{46,47} It will be important to replicate this work with more recent data to see if new career typologies emerge or existing ones change in size or type. Other important shifts are occurring within primary care, particularly the growth of direct primary care and concierge medicine, which may create new career typologies for PCPs but were not yet large enough in number to be reflected in this study.

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